GISS 2026

Photons or Priors

What learned models can and cannot measure, and how to tell the difference before you run anything.

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FIGURE NEEDED — the super-resolved galaxy, full bleed.
Ideally the paired strip: input · model output · second-instrument truth.

THE QUESTION

Same photons. More detail.

Where did the information come from?

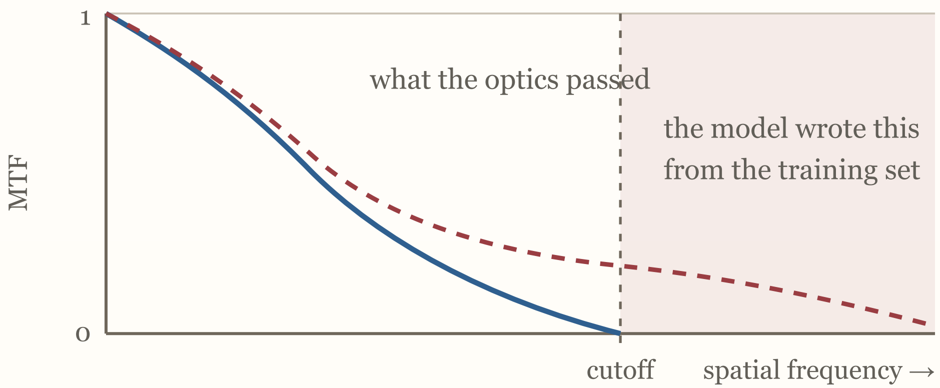
Three of our papers sharpened images. One did not. This talk is about why the difference was predictable in advance.

No process adds information about the sky.

$$Sky \rightarrow Data \rightarrow f(Data)$$
$$I(Sky\ ;\ f(Data)) \leq I(Sky\ ;\ Data)$$

The data processing inequality. No architecture, no amount of compute. Sharpening is **redistribution**, not acquisition.

> *The honest refinement: a learned prior does carry real sky information — but* ***population*** *information, not* ***this-source*** *information. A statement about galaxies in general, applied to your galaxy.*



Past the band limit the data carry nothing. Whatever the output shows there did not come from this source.

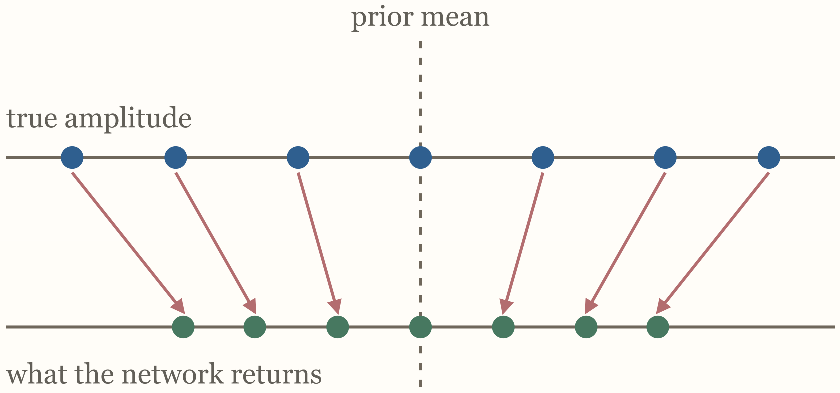
THE MECHANISM

What a prior does to a number

A network trained on pixel-wise error returns an approximation of the **posterior mean**. Posterior means are under-dispersed: they shrink toward the prior.

> *Any per-source amplitude that goes through an MSE-trained network comes back* ***too small*** *— in a direction set by the training distribution.*

A prediction, not a postmortem. Everything after this slide was already implied by it.



The spread collapses toward the mean of whatever the model was trained on.

THE RECEIPTS · THREE PAPERS

Predicted, then measured

PREDICTED, FROM THEORY	MEASURED, IN OUR PAPERS
MSE nets shrink per-source amplitudes	ellipticity shrunk **25–35%** *Everetts+* · median Hα recovered **26%** of true *Haghjoo+* · flux biased **up** by oversmoothing *Rezaee+*
generative priors trade accuracy for realism	the diffusion model suppresses shape information **below bilinear interpolation** *Everetts+*
priors help decisions, hurt measurements	the one clean win is deblending at 3–5″: **35%** vs **9%** peak recovery *Rezaee+* — a decision, auditable against truth
the gap is invisible from inside	every gap above was measured against a **second instrument**: JWST · Spitzer · the grating · truth catalogs

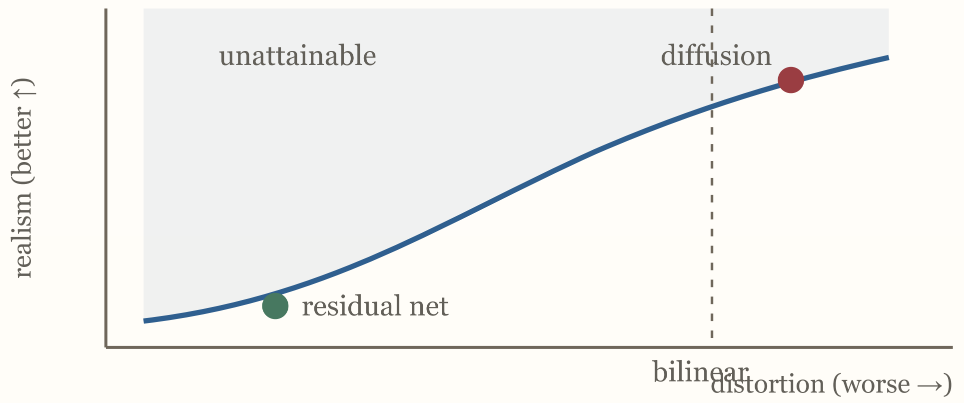
> *Held-out validation from the same data looked fine in every case.*

Realism and accuracy are provably in tension

Past a bound, perceptual quality is bought only by paying distortion *(Blau & Michaeli 2018)*.

> *The prettiest reconstruction is the least faithful one — not sometimes, structurally.*

Scope it aloud: on structural metrics, for this task.



The diffusion model sits to the right of the bilinear baseline: more realistic, and further from the truth.

THE MONDAY-MORNING TOOL

Decisions versus measurements

SAFE — A DECISION

Few bits. Is there a source? Is this one object or two?

Checkable against truth: completeness, purity. A prior that is wrong here shows up in the audit.

UNSAFE — A MEASUREMENT

A continuous number. A shape, a flux, a line ratio.

Propagates silently into cosmology or chemistry, with no per-object truth to catch it.

Never measure on enhanced pixels. Detect on them if you like.

THE AUDIT

You cannot estimate the gap from inside.

Validation is denominated in the same currency as training. The shape bias survived every split and died at JWST.

Same currency

Held-out data share the instrument, the selection, and the prior. They cannot price what all three omit.

Second instrument

JWST for shapes. Spitzer for photometry. The grating for spectra. Different optics, different failure modes.

What it bought

Every number on the receipts slide. None of it was visible without a second channel.

THE FIELD ALREADY KNOWS THIS

Two converged losses, one universe.

Both pipelines converged. Both report tight error bars. They disagree.

The field's answer was not a better optimizer. It was more independent channels.

ACT II

A catalog is a lossy codec.

Petabytes to kilobytes, with the distortion measure chosen in advance.

> *The columns are the distortion measure. You decide which questions stay answerable before you know what will be asked.*

Which is a fine bargain, provided someone says out loud what was thrown away.

THE MECHANISM

What do ten instruments share?

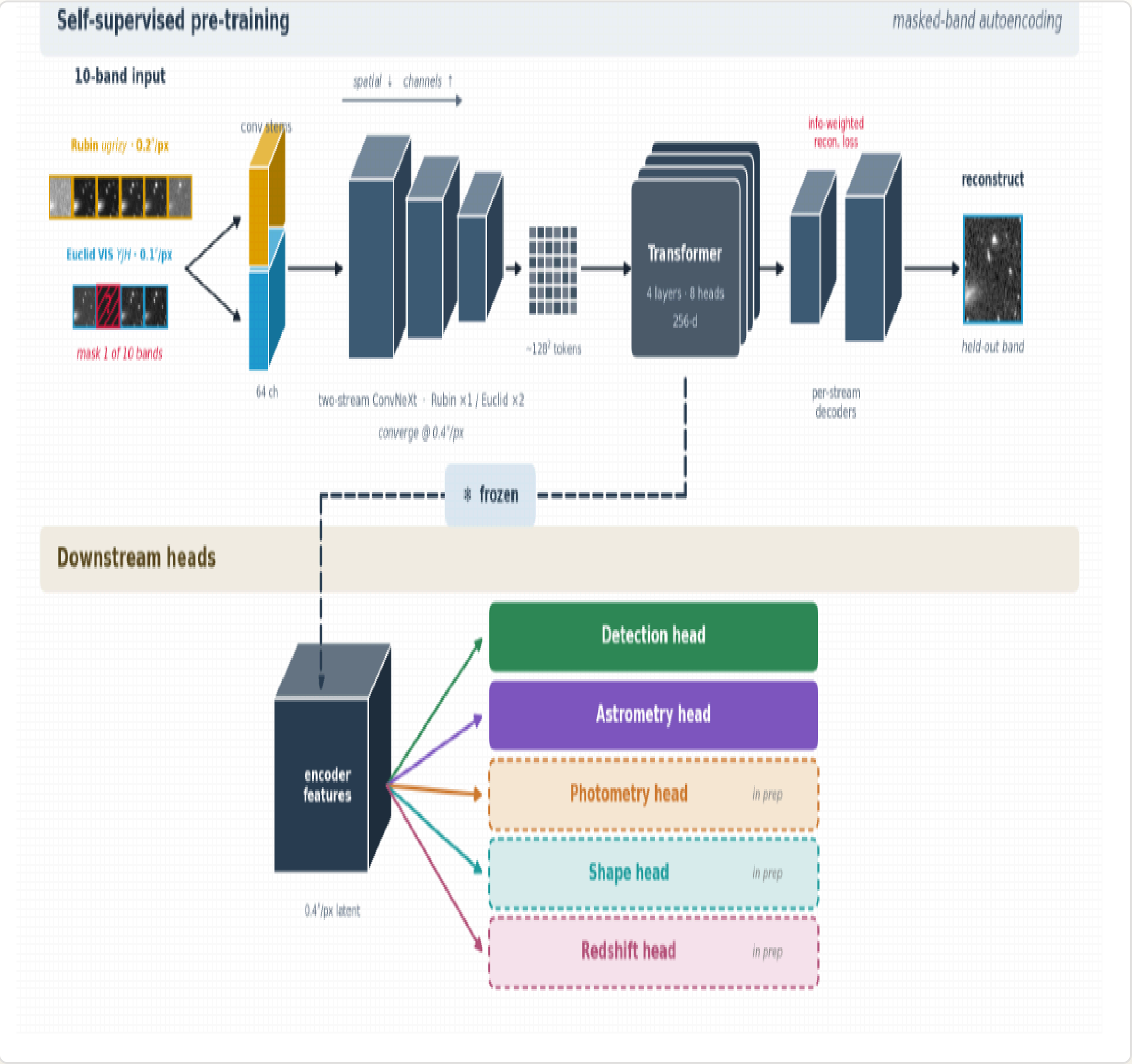
mask one band · predict it from the other nine

No band is predictable from the other nine unless the model separates the source from what each instrument did to it.

> *The only thing ten instruments share is the sky. So the factorization is not a bonus — it is forced.*

Alignment is not an add-on to the reconstruction. It is the only way to score on the task.

Ten bands, no common grid, no resampling.



Rubin ugrizy + Euclid VIS·Y·J·H → two-stream stems → shared latent → heads. ~9M parameters, self-supervised, each band at its delivered sampling.

- Detection matches the VIS catalog at **93%** complete / **94%** pure — and reaches **0.45 mag** deeper.
- Astrometric scatter **50 → 14–17 mas**.
- Transfers to a new field with **zero retraining**.

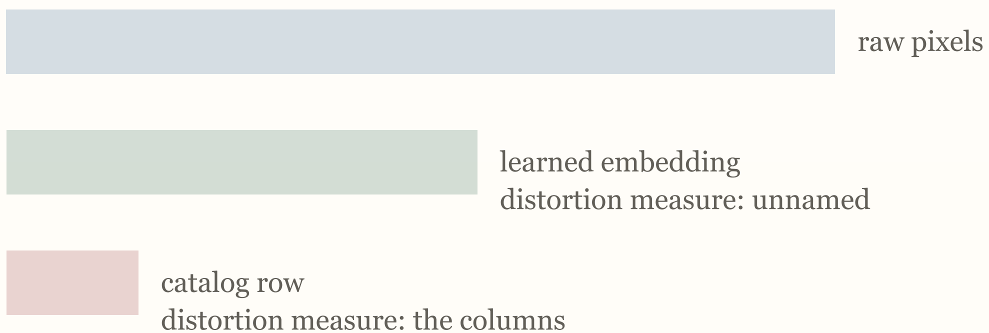
> *Audited in three layers, so the claim is checkable: the **output** (reconstruct the withheld band), the **frozen representation** (the heads only ever see the frozen encoder), and **reuse** (a small new head, on a task it was never trained for).*

THE CONFESSION

Hedging costs bits.

Refusing to name the distortion measure means you cannot claim the catalog's compression ratio. JAISP compresses the **survey**, not the **source**.

Three kinds of missing information: **never observed** · **observed and discarded** · **retained but inaccessible to the model you chose**.



bar length is schematic — not measured bytes

THE FOUR AUDIT RULES

Photon or prior. Say which.

— Detect on enhanced pixels if you like. Do not measure on them.

— Audit against a second instrument. You cannot price the gap from inside.

— Name what the prior contributed: population information, not this-source information.

— Name the distortion measure — or do not claim the compression.

Every pixel that looks new came from a photon or from a prior.